Homotopical Encryption: Hidden Hodge Operators and the Harmonic Cochain Problem

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Abstract

We introduce the Harmonic Cochain Problem (HCP): recovering the operative harmonic component $h = P_{\ker(\Delta_1)}b$ from public data (α, b). The degree-one Hodge–Laplacian Δ_1 is induced by an operative hidden representative $x \in A_\alpha = \pi^{-1}(\alpha)$ of the public classifier value α. Its adjacent coboundaries δ_0, δ_1 determine the Hodge decomposition $b = h + e$, with $e \in \mathrm{im}(\delta_0) \oplus \mathrm{im}(\delta_1^*)$. We compare HCP with the closest-vector problem (CVP), distinguishing the field-linear image $\mathcal{V}_B(R) = \{Bx : x \in R^n\}$ from the discrete lattice $\Lambda_B = B\mathbb{Z}^n$. The former can be realized as a linear kernel. For an explicitly supplied inclusion $\Lambda_B \subseteq W$, orthogonal projection onto the harmonic subspace W preserves the closest lattice vectors. We then prove a hidden-operator separation: if publicly equivalent hidden operators disagree on harmonicity, a correct known-lattice CVP oracle cannot supply a uniformly exact public-data HCP solver.

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1 Introduction

1.1 Motivation and Contributions

Helfer's first-order homotopical logic [Hel25] provides the original semantic motivation for this framework: equality is interpreted through path objects and, under the stated hypotheses, closed-formula truth is invariant under homotopy equivalence of interpretations. This motivates distinguishing a hidden presentation from the semantic information retained by a public classifier. HCP asks how that distinction interacts with the harmonic target determined by a hidden Hodge operator.

Homotopical Encryption studies the recovery of an operative harmonic component when the Hodge operator is hidden. The public input specifies an observed cochain and a class of possible private presentations. In a known-lattice closest-vector problem, the lattice is presented before the search begins. In HCP, the recovery target is

$$h = P_H(b) \in \ker(\Delta_1),$$

but the public class does not include the degree-one Hodge–Laplacian Δ_1, its kernel, or the restriction maps from which Δ_1 is built.

The operator dependence can be summarized as follows. A hidden representative determines the restriction data, which determine the adjacent coboundaries and hence the degree-one Hodge operator:

$$x \longmapsto \{R^x_{\sigma\to\tau}\}_{\sigma\prec\tau} \longmapsto (\delta_{0,x}, \delta_{1,x}) \longmapsto \Delta_{1,x} \longmapsto \big(\ker \Delta_{1,x}, P_{H_x}\big).$$

Homotopy-equivalent hidden representatives share a public label even when their Hodge operators differ in the fixed public coordinates.

The CVP comparison makes this dependence on the input precise. A real linear span can be realized as the kernel of a suitable linear operator, but a discrete lattice cannot equal a nonzero real kernel. If an explicit lattice Λ is supplied inside a harmonic subspace $W = \ker(\Delta_W)$, orthogonal projection of the target onto W preserves the closest vectors of Λ.

Symbol	Meaning
$\pi : A \to P$	homotopy-invariant public classifier
$\alpha = \pi(x) \in P$	realized public classifier value
$A_\alpha = [x]_\pi$	fiber of hidden representatives with public label α
(α, b)	public HCP observation
$a \in C^0,\ c = \delta_0 a \in C^1$	degree-zero opening and committed exact one-cochain
$\delta_0, \delta_1, \Delta_1$	adjacent coboundaries and degree-one Hodge–Laplacian
$\ker(\Delta_1),\ P_H$	harmonic subspace and harmonic projection
$\Delta_W = I - P_W$	kernel-normalized projection operator associated with W

Table 1: Core notation for the semantic interpretation and HCP.

This paper develops this boundary through the following statements.

(i) The Harmonic Cochain Problem (HCP) asks to recover the operative component $h = P_H b$ for a hidden Hodge operator Δ_1.

(ii) A Hodge-structured noise model gives the cochain decomposition $b = h + e$, where $h \in \ker(\Delta_1)$ and $e \in \mathrm{im}(\delta_0) \oplus \mathrm{im}(\delta_1^*)$.

(iii) Harmonic lattices in one public class can have different unique closest vectors to the same cochain, so known-lattice CVP access does not determine the operative answer.

(iv) Exact-component extraction can lose the full Hodge residual.

Finite Hodge theory follows the cellular-cochain framework of Hansen–Ghrist [HG19]; the algebraic model used here is specified in Section 2. The constructed public theories use the nonzero-cohomology-test fragment.

2 The Harmonic Cochain Problem

Let A be the type of hidden presentations, with semantic interpretation M_x for each $x \in A$. Write $x \simeq y$ for homotopy equivalence of M_x and M_y in Helfer's sense [Hel25, Definition 7.1]. Assume that this equivalence preserves the chosen closed tests. Let P be the type of public class data and choose a classifier $\pi : A \to P$ that factors through their closed theory. Then

$$x \simeq y \quad \Longrightarrow \quad \pi(x) = \pi(y).$$

The typed data and the invariance premise are made precise in Definition 2.12 and Proposition 2.13. For a realized public value α, write

$$\alpha \in \mathrm{im}(\pi), \qquad A_\alpha = \pi^{-1}(\alpha) = \{x \in A : \pi(x) = \alpha\}.$$

For $x \in A$, we also write $[x]_\pi := A_{\pi(x)}$. Thus, whenever $\alpha = \pi(x)$, the two descriptions agree: $A_\alpha = [x]_\pi$. We call the classifier *complete for the declared homotopy equivalence* when $\pi(x) = \pi(y)$ implies $x \simeq y$ for all $x, y \in A$.

We attach Hodge data to each presentation using finite graded cell carriers, integer incidence numbers and face-to-coface linear stalk maps.

Fix a finite coordinate set N and let $V = \mathbb{R}^N$ have its Euclidean inner product. Each presentation has finite-dimensional real inner-product cochain spaces C^0_x, C^1_x, C^2_x. To compare presentations using one public cochain vector, fix isometric coordinate charts $\iota_x : C^1_x \xrightarrow{\sim} V$. We identify degree-one vectors and maps with their transported versions in V; the charts are part of the comparison data.

The restriction maps $R^x_{\sigma\to\tau}$ for $\sigma \prec \tau$, including vertex-to-edge and edge-to-face maps, determine adjacent coboundaries

$$\delta_{0,x} : C^0_x \to V, \qquad \delta_{1,x} : V \to C^2_x, \qquad \delta_{1,x}\delta_{0,x} = 0.$$

In these coordinates, define the degree-one Hodge–Laplacian, its harmonic subspace and its orthogonal projector by

$$\begin{aligned} \Delta_x &= \delta_{0,x}\delta^*_{0,x} + \delta^*_{1,x}\delta_{1,x}, \\ \mathcal{H}_x &= \ker(\Delta_x), \qquad P_{\mathcal{H}_x} : V \to \mathcal{H}_x \subseteq V. \end{aligned}$$

The stars denote adjoints. When the operative representative is fixed, we suppress x and also write Δ_1 or Δ for its degree-one Laplacian, and P_H for its harmonic projector.

In the Homotopical Encryption public-sample model, a presentation x produces a cochain

$$b = h + e, \qquad h \in \mathcal{H}_x, \qquad e \in \mathrm{im}(\delta_{0,x}) \oplus \mathrm{im}(\delta^*_{1,x}).$$

The message h is a harmonic representative of its degree-one cohomology class, and e is Hodge-structured noise. When x is also equipped with an explicit harmonic-lattice presentation, write $\Lambda_x \subseteq \mathcal{H}_x$ for its discrete lattice and choose $h \in \Lambda_x$. In this integral profile, the public fiber indexes a family of possible embedded harmonic lattices; the operative hidden member selects one of them.

Definition 2.1 (Public observation)**.** *A public HCP observation is a pair* (α, b) *consisting of a realized public classifier value* $\alpha \in \mathrm{im}(\pi)$ *and an observed degree-one cochain* $b \in V$ *in the fixed comparison coordinates.*

Definition 2.2 (Hidden Hodge witness)**.** *A hidden Hodge witness for a public observation* (α, b) *is a hidden representative* $x \in A_\alpha$*, satisfying* $\pi(x) = \alpha$*. Its presentation data select the operative operator* Δ_x*. The witness and* b *determine the coordinates*

$$h_x(b) := P_{\mathcal{H}_x} b, \qquad e_x(b) := b - h_x(b).$$

The Hodge decomposition in Proposition 2.7 gives

$$h_x(b) \in \ker(\Delta_x), \qquad e_x(b) \in \mathrm{im}(\delta_{0,x}) \oplus \mathrm{im}(\delta_{1,x}^*), \qquad b = h_x(b) + e_x(b).$$

Definition 2.3 (Harmonic Cochain Problem)**.** *The* Harmonic Cochain Problem *(HCP) asks, given the public observation* (α, b)*, to recover the operative harmonic component*

$$h = h_x(b) \in \mathcal{H}_x,$$

equivalently the harmonic term in the orthogonal Hodge decomposition of b *for the operative hidden witness* $x \in A_\alpha$*. The witness is bound to the instance but is not part of the public input.*

The algebraic cellular model above adapts the finite Hodge setup of [HG19] without assuming a regular CW or simplicial realization for every admitted incidence structure. Such a realization and the separate simplicial-coherence conditions require additional data.

Definition 2.4 (Matrix harmonicity)**.** *For a hidden linear Hodge operator* Δ*, a cochain* $h \in \mathbb{R}^N$ *is* Δ*-harmonic when*

$$\Delta h = 0.$$

Thus the harmonic kernel is $\ker(\Delta)$*. In the projection model this agrees with the condition* $h \in W$ *for the kernel-normalized operator* $\Delta_W = I - P_W$*.*

Definition 2.5 (Kernel-normalized operator, harmonic projection, and residual)**.** *Let* $V = \mathbb{R}^n$ *be equipped with its standard inner product, and let* $W \subset V$ *be a linear subspace. Write*

$$P_W : V \to W \subset V \qquad \textit{and} \qquad P_{W^\perp} : V \to W^\perp \subset V$$

for the orthogonal projections. The kernel-normalized projection operator attached to W is

$$\Delta_W = I - P_W.$$

For $b \in V$, define the residual vector and residual score by

$$r_W(b) = \Delta_W b, \qquad \rho_W(b) = \|r_W(b)\|.$$

We say that b is W-harmonic when $r_W(b) = 0$. This operator records exactly the harmonic kernel and its orthogonal projection. It is not asserted to equal a cellular Hodge–Laplacian $\delta_0\delta_0^* + \delta_1^*\delta_1$, or to have the same positive spectrum, unless the cellular Laplacian additionally acts as the identity on $W^\perp$.

Proposition 2.6 (Projection harmonicity equivalences)**.** *For every $b \in V$,*

$$r_W(b) = P_{W^\perp} b, \qquad \rho_W(b) = \|P_{W^\perp} b\|, \qquad \rho_W(b) \geq 0.$$

Moreover

$$P_W b \in W, \qquad r_W(P_W b) = 0,$$

and the following conditions are equivalent:

$$r_W(b) = 0 \iff \rho_W(b) = 0 \iff b \in W.$$

Proof. Since $V = W \oplus W^\perp$ orthogonally, every $b \in V$ decomposes uniquely as

$$b = P_W b + P_{W^\perp} b.$$

Hence

$$(I - P_W)b = b - P_W b = P_{W^\perp} b,$$

giving $r_W(b) = P_{W^\perp} b$ and the displayed formula for $\rho_W(b)$. Nonnegativity is the nonnegativity of the norm. The projection $P_W b$ lies in W, so its orthogonal component is zero; therefore $r_W(P_W b) = 0$. Finally, $r_W(b) = 0$ iff $P_{W^\perp} b = 0$, which holds iff $b \in W$. The equivalence with $\rho_W(b) = 0$ follows from $\|x\| = 0$ iff $x = 0$. □

Proposition 2.7 (Hodge decomposition surface)**.** *Let the three consecutive cochain spaces be finite-dimensional real inner product spaces, and let the displayed stars denote adjoints. Suppose*

$$d_{\text{next}} d_{\text{prev}} = 0, \qquad \Delta = d_{\text{prev}} d_{\text{prev}}^* + d_{\text{next}}^* d_{\text{next}}.$$

Then every cochain b admits a decomposition

$$b = h + e_1 + e_2$$

with

$$\Delta h = 0, \qquad e_1 \in \mathrm{im}(d_{\mathrm{prev}}), \qquad e_2 \in \mathrm{im}(d^*_{\mathrm{next}}),$$

and

$$\langle h, e_1 \rangle = \langle h, e_2 \rangle = \langle e_1, e_2 \rangle = 0.$$

The decomposition is unique.

Proof. Each summand defining Δ is self-adjoint, so Δ is self-adjoint. In finite dimension, $\mathrm{im}(\Delta) = (\ker \Delta)^{\perp}$, and hence the ambient cochain space decomposes as

$$\ker(\Delta) \oplus \mathrm{im}(\Delta).$$

Since

$$\mathrm{im}(\Delta) \subseteq \mathrm{im}(d_{\mathrm{prev}}) + \mathrm{im}(d^*_{\mathrm{next}}),$$

every cochain b can be written as $h + e_1 + e_2$ with $h \in \ker \Delta$, $e_1 \in \mathrm{im}(d_{\mathrm{prev}})$, and $e_2 \in \mathrm{im}(d^*_{\mathrm{next}})$. From $\Delta h = 0$, one obtains

$$\|d^*_{\mathrm{prev}} h\|^2 + \|d_{\mathrm{next}} h\|^2 = 0,$$

hence $d^*_{\mathrm{prev}} h = 0$ and $d_{\mathrm{next}} h = 0$. These identities give $\langle h, e_1 \rangle = 0$ and $\langle h, e_2 \rangle = 0$, while $d_{\mathrm{next}} d_{\mathrm{prev}} = 0$ gives $\langle e_1, e_2 \rangle = 0$. Hence the harmonic, exact, and coexact subspaces are pairwise orthogonal. Their sum is therefore direct, which makes the three components unique. □

2.1 Hodge Encryption and Correctness

For a hidden representative x, define its noise space by

$$\mathcal{E}_x = \mathcal{H}_x^{\perp}.$$

The Hodge decomposition identifies the noise space as the orthogonal direct sum

$$\mathcal{E}_x = \mathrm{im}(\delta_{0,x}) \oplus^{\perp} \mathrm{im}(\delta^*_{1,x}),$$

and gives the orthogonal decomposition

$$C^1_x = \mathcal{H}_x \oplus^{\perp} \mathcal{E}_x.$$

Consequently,

$$\begin{aligned} \mathrm{combine}_x : \mathcal{H}_x \times \mathcal{E}_x \longrightarrow C^1_x, \qquad & (h, e) \longmapsto h + e, \\ \mathrm{split}_x : C^1_x \longrightarrow \mathcal{H}_x \times \mathcal{E}_x, \qquad & b \longmapsto \left(P_{\mathcal{H}_x} b, P_{\mathcal{E}_x} b\right) \end{aligned}$$

are inverse linear bijections. Orthogonality also gives

$$\|h+e\|^2 = \|h\|^2 + \|e\|^2.$$

This bijection is the algebraic encryption/decryption construction. Given a harmonic message $h \in \mathcal{H}_x$, randomized Hodge encryption samples, or otherwise chooses, admissible noise $e \in \mathcal{E}_x$ and sets

$$\mathsf{Enc}_x(h;e) = \mathrm{combine}_x(h,e) = h+e.$$

Full coordinate recovery is split_x; message decryption retains its harmonic coordinate:

$$\mathsf{Dec}_x(b) = \mathrm{pr}_{\mathcal{H}_x}\big(\mathrm{split}_x(b)\big) = P_{\mathcal{H}_x} b.$$

For every admissible noise value,

$$\mathsf{Dec}_x\big(\mathsf{Enc}_x(h;e)\big) = h, \qquad \mathrm{split}_x\big(\mathrm{combine}_x(h,e)\big) = (h,e).$$

The bijection concerns the full message–noise pair (h,e). Choosing a noise distribution makes Enc_x randomized, while Dec_x recovers the message for every admissible noise value. Both maps depend on the operative representative x.

Exact-component observations. Fix one cellular Hodge system and an opening $a \in C^0$. Its exact one-cochain $c = \delta_0 a$ satisfies

$$P_H c = P_H \delta_0 a = 0.$$

Indeed, $\mathrm{im}\,\delta_0$ is orthogonal to the harmonic subspace. Thus this exact cochain is not the full sample $b = h+e$ when the harmonic message h is nonzero.

There is also a distinction between the exact one-cochain and its opening. Linearity gives

$$\delta_0 a = \delta_0 a' \quad \Longleftrightarrow \quad a - a' \in \ker \delta_0.$$

Consequently the raw coboundary determines an opening only modulo $\ker \delta_0$. A unique degree-zero message encoding requires further encoding data.

Proposition 2.8 (Exact-component obstruction)**.** *Suppose $v \in C^2$ and $\delta_1^* v \neq 0$. On the unrestricted Hodge-noise carrier, the samples*

$$b_t = t\delta_1^* v, \qquad t \in \mathbb{R},$$

have zero harmonic message and zero exact component, while

$$\|P_{\mathcal{H}^\perp} b_t\| = |t|\,\|\delta_1^* v\|.$$

Thus this full-observation residual does not factor through the exact component. If an extraction depends only on that component and its zero image has target distance zero, it cannot preserve any positive target far radius uniformly on this unrestricted carrier, at any source radius.

Proof. A coexact vector is orthogonal to $\mathcal{H}$, so its orthogonal projection onto $\mathcal{H}^{\perp}$ is itself. Homogeneity gives the displayed norm. The exact-component view is constantly zero, whereas this norm is zero at $t = 0$ and positive at $t = 1$, proving nonfactorization. For any fixed source radius, choose positive t large enough to exceed it. The extracted value still has distance zero, contradicting every positive target far radius. □

A restricted noise or realization profile may exclude this family, so a comparison for such a profile requires its explicit hypotheses. The scores

$$\|P_{H^{\perp}} b\|, \qquad \operatorname{dist}(b, \operatorname{im} \delta_0), \qquad \text{a modular cochain or code distance}$$

measure harmonic residual, distance to exactness, and modular or code disagreement, respectively. A modular distance needs its own carrier and normalization.

2.2 Public and Hidden Data

A record presenting the public observation must disclose b, or specify its extraction, and bind it to the operative operator, classifier and sample. An exact-component payload or a proof about that component alone does not supply the full cochain. Equal decoded components also need not imply equal raw payload bytes.

For a larger released view $\mathsf{View}(x, b)$, an ambiguity argument requires equality of that view. Factoring through (α, b) suffices; otherwise equality must be established for the actual released fields.

Definition 2.9 (Hidden-operator public projection)**.** *For a public class type P, a public projection is a map*

$$\pi_{\Delta} : \{\textit{hidden Hodge operators } \Delta\} \longrightarrow P.$$

Two hidden operators are publicly equivalent when $\pi_{\Delta}(\Delta) = \pi_{\Delta}(\Delta')$. The hidden operator fiber over $\alpha \in P$ is

$$\operatorname{Fib}_{\pi_{\Delta}}(\alpha) = \pi_{\Delta}^{-1}(\alpha) = \{\Delta : \pi_{\Delta}(\Delta) = \alpha\},$$

and the hidden class of Δ is $\operatorname{Fib}_{\pi_{\Delta}}(\pi_{\Delta}(\Delta))$.

Definition 2.10 (Lossiness and kernel ambiguity)**.** *The public projection* π_Δ *is* lossy *if there exist* $\Delta \neq \Delta'$ *with* $\pi_\Delta(\Delta) = \pi_\Delta(\Delta')$*. The two operators have the same harmonic kernel if*

$$\forall v, \quad \Delta v = 0 \Longleftrightarrow \Delta' v = 0.$$

The projection is kernel-ambiguous *if there are publicly equivalent operators* Δ, Δ' *whose harmonic kernels are not the same.*

Definition 2.11 (Public-compatible Hodge ambiguity)**.** *Public-compatible Hodge ambiguity means that there exist publicly equivalent operators* Δ, Δ' *and a cochain-facing observation* b *such that*

$$\Delta b = 0 \qquad \textit{but} \qquad \Delta' b \neq 0.$$

Definition 2.12 (Typed HCP semantic realization)**.** *Fix a multi-sorted algebraic signature* σ*, a model category* $\mathcal{C}$ *with terminal object and binary products, and, for each hidden representative* $x \in A$*, a base interpretation* $M_x : \sigma \to \mathcal{C}_{\mathrm{f}}$ *together with homotopical formula semantics over* M_x*. Write*

$$\mathrm{Th}(M_x) = \{\varphi \textit{ closed} : M_x \vDash \varphi\}$$

for its closed first-order theory. Relative to the presentation data and coordinate charts fixed above, a typed HCP semantic realization consists additionally of maps

$$\begin{aligned} x &\longmapsto \Delta_x \in \mathrm{End}(V), \\ \kappa &: \{\textit{closed theories}\} \longrightarrow P, \\ \pi_\Delta &: \{\textit{Hodge operators}\} \longrightarrow P, \end{aligned}$$

satisfying the two factorization identities

$$\pi(x) = \kappa(\mathrm{Th}(M_x)), \qquad \pi_\Delta(\Delta_x) = \pi(x).$$

The Laplacian Δ_x *is presentation-level Hodge data extending the semantic core* M_x*; it is not silently included among the symbols whose closed theory is asserted to be homotopy invariant.*

Proposition 2.13 (Helfer-to-HCP classifier bridge)**.** *Let* $x, y \in A$ *belong to a typed HCP semantic realization. If* M_x *and* M_y *are homotopy equivalent interpretations in the sense of Helfer, then*

$$\pi(x) = \pi(y) \qquad \textit{and} \qquad \pi_\Delta(\Delta_x) = \pi_\Delta(\Delta_y).$$

If, independently, there is a cochain b for which

$$\Delta_x b = 0, \qquad \Delta_y b \neq 0,$$

then π_Δ has public-compatible Hodge ambiguity, and hence is kernel-ambiguous and lossy.

Proof. The stated closed-test invariance premise gives, for every closed σ-formula φ,

$$M_x \vDash \varphi \quad \Longleftrightarrow \quad M_y \vDash \varphi.$$

In the full first-order setting this premise is supplied by Helfer's homotopy-invariance theorem [Hel25, Theorem 12.3]. Hence $\mathrm{Th}(M_x) = \mathrm{Th}(M_y)$. Applying κ and then the two factorization identities gives

$$\pi(x) = \pi(y), \qquad \pi_\Delta(\Delta_x) = \pi_\Delta(\Delta_y).$$

The additional displayed harmonicity and non-harmonicity conditions now exhibit (Δ_x, Δ_y, b) as a public-compatible Hodge-ambiguity witness. The ambiguity hierarchy below supplies kernel ambiguity and lossiness. □

Under closed-test invariance and classifier completeness, Proposition 2.13 gives

$$\pi(x) = \pi(y) \quad \Longleftrightarrow \quad x \simeq y, \qquad A_{\pi(x)} = \{y \in A : y \simeq x\}.$$

The implication from label equality uses completeness; the other uses invariance. The fiber identity follows from the definition of $A_{\pi(x)}$. The family in Proposition 3.5 has a complete classifier. Completeness concerns the semantic class; it does not select its operative embedded harmonic kernel. Kernel disagreement remains an independent condition. If Δ-harmonicity were included in the invariant first-order signature, and the common coordinate b were named compatibly in both interpretations, Helfer's theorem would instead rule out the displayed disagreement.

Suppose an equivalence of hidden presentations induces degreewise linear isometric equivalences U_0, U_1, U_2 satisfying

$$U_1 \delta_{0,x} = \delta_{0,y} U_0, \qquad U_2 \delta_{1,x} = \delta_{1,y} U_1.$$

In the fixed coordinate charts, U_1 is an isometry of V. The isometries transport adjoints, giving

$$\Delta_y U_1 = U_1 \Delta_x, \qquad U_1(\ker \Delta_x) = \ker \Delta_y.$$

The corresponding orthogonal projections commute with transport:

$$U_1 P_{\ker \Delta_x} b = P_{\ker \Delta_y} U_1 b.$$

Thus a transported observation preserves both the operative target and harmonicity:

$$\Delta_x b = 0 \quad \Longleftrightarrow \quad \Delta_y(U_1 b) = 0.$$

The ambiguity witness in Proposition 2.13 uses a fixed public coordinate $b \in V$. Under the displayed conjugacy, its separation condition is equivalently

$$b \in \ker \Delta_x, \qquad b \notin U_1(\ker \Delta_x).$$

It requires a nonstabilization witness in addition to the equivalence, with the private transport U_1 neither revealed nor applied.

Proposition 2.14 (Ambiguity hierarchy)**.** *Public-compatible Hodge ambiguity implies kernel ambiguity, and kernel ambiguity implies that the public projection is lossy.*

Proof. If b is harmonic for Δ but not for Δ', then the two operators cannot have the same harmonic kernel. This gives kernel ambiguity. If publicly equivalent operators have different kernels, they cannot be the same operator, since equality of operators implies equality of kernels. Therefore the public projection identifies two distinct hidden operators. □

Definition 2.15 (HCP solution)**.** *Relative to an operative hidden witness x, a solution to an HCP instance (α, b) is the cochain $h' = h_x(b) = P_{\ker(\Delta_x)} b$. A public-data-only solver is* uniformly exact *if the same output has this value for every hidden witness compatible with its public input.*

2.3 Computational Scope

These exact-recovery comparisons do not by themselves give an asymptotic CVP-hardness reduction. Such a comparison requires encoded instance families, an input-size measure, an algorithm class and a success criterion. A distributional security claim also requires a sampling model and a declared adversary view.

3 CVP Comparison and Kernel Ambiguity

CVP asks for a closest vector in an explicitly presented lattice [MG02]. HCP asks for the projection of b into the kernel of the operative hidden Hodge operator.

3.1 Known-Lattice CVP

An explicit Euclidean CVP instance consists of a full-column-rank generator matrix $B \in \mathbb{R}^{m \times n}$ and a target $t \in \mathbb{R}^m$. Its discrete lattice is

$$\Lambda_B = B\mathbb{Z}^n = \{Bz : z \in \mathbb{Z}^n\}.$$

The injectivity of $z \mapsto Bz$ records that the displayed columns are a lattice basis. A correct closest-vector answer is a vector $v \in \Lambda_B$ such that

$$\|t - v\|^2 \leq \|t - w\|^2 \qquad \text{for every } w \in \Lambda_B.$$

Thus, for fixed dimensions, a CVP oracle O is correct when

$$\begin{aligned} \mathsf{CVP}_{\mathrm{corr}}(O) \quad \Longleftrightarrow \quad & \forall \text{ full-column-rank } B, t, \\ & O(B,t) \in \Lambda_B \quad \text{and} \quad \forall w \in \Lambda_B, \\ & \|t - O(B,t)\|^2 \leq \|t - w\|^2. \end{aligned}$$

The weaker output condition is merely lattice membership:

$$\forall B, t,\ O(B,t) \in \Lambda_B.$$

Fix a class $\mathcal{C}$ of admissible algorithms, such as polynomial-time or probabilistic polynomial-time algorithms. The external CVP hardness assumption is the schema

$$\mathsf{CVP}_{\mathrm{hard}}(\mathcal{C}) \quad \Longleftrightarrow \quad \neg\exists O,\ O \in \mathcal{C} \wedge \mathsf{CVP}_{\mathrm{corr}}(O).$$

3.2 Presented Linear Images

Let R be a field. The field-linear image of B is a different object:

$$\mathcal{V}_B(R) = BR^n = \{Bx : x \in R^n\}.$$

Over a field this is a linear subspace, not the discrete lattice $B\mathbb{Z}^n$.

Let B present $\mathcal{V}_B(R)$. If a matrix Δ satisfies $\ker(\Delta) = \mathcal{V}_B(R)$, then any solver satisfying

$$\Delta S(\Delta, t) = 0$$

returns $S(\Delta, t) \in \mathcal{V}_B(R)$.

Proposition 3.1 (Kernel realization for explicit generated subspaces)**.** *Let $B \in R^{n \times n}$ be an explicit square generator matrix over a field and set $\mathcal{V}_B(R) = \{Bx : x \in R^n\}$. There exists a matrix Δ such that*

$$\ker(\Delta) = \mathcal{V}_B(R).$$

Consequently, any solver $S(\Delta, b)$ whose output is Δ-harmonic for every presented Δ returns a vector in the explicit generated subspace $\mathcal{V}_B(R)$ when run on this realized operator.

Proof. Let $B \in R^{n\times n}$ be an explicit generator matrix over a field R, set

$$\mathcal{V}_B(R) = \{Bx : x \in R^n\} \subseteq R^n,$$

and let $t \in R^n$ be the solver input. We construct a matrix Δ with

$$\ker(\Delta) = \mathcal{V}_B(R).$$

Put $V = \mathcal{V}_B(R)$. Since R^n is finite-dimensional over a field, choose a complementary subspace W with

$$V + W = R^n, \qquad V \cap W = 0.$$

Thus every $y \in R^n$ has a unique decomposition

$$y = v + w, \qquad v \in V, \quad w \in W.$$

Equivalently, there is a unique element $w_y \in W$ such that $y - w_y \in V$. Define

$$P(y) = w_y.$$

The uniqueness of the $V \oplus W$ decomposition gives linearity: for $y, z \in R^n$, the vector $P(y) + P(z)$ lies in W, and

$$(y + z) - (P(y) + P(z)) = (y - P(y)) + (z - P(z)) \in V,$$

so $P(y + z) = P(y) + P(z)$. Similarly, for $a \in R$, $aP(y) \in W$ and

$$ay - aP(y) = a(y - P(y)) \in V,$$

so $P(ay) = aP(y)$.

We now compute the kernel. If $y \in V$, then $0 \in W$ and $y - 0 \in V$, hence uniqueness gives $P(y) = 0$. Conversely, if $P(y) = 0$, then $y - P(y) = y \in V$. Therefore

$$\ker P = V = \mathcal{V}_B(R).$$

Let Δ be the matrix of the linear map P. Then, for every $y \in R^n$,

$$\Delta y = 0 \quad \Longleftrightarrow \quad y \in \ker P \quad \Longleftrightarrow \quad y \in \mathcal{V}_B(R) \quad \Longleftrightarrow \quad \exists x,\ Bx = y.$$

Let $S : R^{n\times n} \times R^n \to R^n$ be a presented-operator Hodge solver satisfying

$$\forall \Delta\, b, \qquad \Delta(S(\Delta, b)) = 0.$$

Applying this to the Δ constructed above and $b = t$ gives

$$\Delta(S(\Delta, t)) = 0.$$

By the kernel calculation, there exists x with

$$Bx = S(\Delta, t).$$

Hence $S(\Delta, t) \in \mathcal{V}_B(R)$. □

When $R = \mathbb{R}$, if a presented-operator solver additionally minimizes Euclidean distance among Δ-harmonic vectors, the same kernel equality says that it returns the continuous closest point of $\mathcal{V}_B(\mathbb{R})$. This is still not discrete CVP unless a separate lattice is supplied.

3.3 Explicit Harmonic-Lattice Bridge

The discrete bridge requires both kinds of data: a real harmonic subspace and an integer lattice contained in it. Let $W \subseteq \mathbb{R}^m$ be the operative harmonic subspace, let P_W be its orthogonal projection, and let $\Lambda_B = B\mathbb{Z}^n \subseteq W$ be an explicitly presented lattice. Set

$$\Delta_W = I - P_W.$$

The inclusion $\Lambda_B \subseteq W$ does not identify the discrete lattice with the real subspace.

Proposition 3.2 (Discrete CVP through harmonic projection)**.** *For $t \in \mathbb{R}^m$, put $t_W = P_W t$. Then:*

(i) t_W and every $\lambda \in \Lambda_B$ are Δ_W-harmonic;

(ii) for every $\lambda \in \Lambda_B$,

$$\|t - \lambda\|^2 = \|t - t_W\|^2 + \|t_W - \lambda\|^2;$$

(iii) a lattice vector λ is closest to t in Λ_B if and only if it is closest to t_W in Λ_B.

Proof. Since $t_W \in W$, every $\lambda \in \Lambda_B$ also lies in W, and $\ker(\Delta_W) = W$, the harmonicity claims follow. For the distance identity, write

$$t - \lambda = (t - t_W) + (t_W - \lambda).$$

Orthogonal projection gives $t - t_W \in W^\perp$, while $t_W - \lambda \in W$. The two summands are orthogonal, so the Pythagorean identity gives (ii). The term $\|t - t_W\|^2$ is independent of λ; consequently, for any $\lambda, \mu \in \Lambda_B$,

$$\|t - \lambda\|^2 \leq \|t - \mu\|^2 \quad \iff \quad \|t_W - \lambda\|^2 \leq \|t_W - \mu\|^2.$$

This is exactly (iii). □

The continuous projection $t_W = P_W t$ is the closest harmonic vector to t, but it need not lie in Λ_B. Thus projection transports the target to the harmonic subspace while preserving the discrete minimizers; it does not perform the remaining lattice search.

3.4 Oracle Insufficiency

Fix a typed HCP semantic realization as in Definition 2.12. The solver contracts below range over its admitted representatives.

Definition 3.3 (Public-data Hodge solver contract)**.** *Let*

$$S : \mathcal{O}_{\mathrm{CVP}} \times (P \times \mathbb{R}^n) \longrightarrow \mathbb{R}^n,$$

where $\mathcal{O}_{\mathrm{CVP}}$ is the class of explicit-lattice CVP oracles. For a realization $x \mapsto (M_x, \Delta_x)$ and $O \in \mathcal{O}_{\mathrm{CVP}}$, define

$$\begin{aligned} \mathsf{H}_O(S) &\iff \forall x : A, \quad \forall b \in \mathbb{R}^n, \quad \Delta_x\big(S(O, (\pi(x), b))\big) = 0, \\ \mathsf{F}_O(S) &\iff \forall x : A, \quad \forall b \in \mathbb{R}^n, \quad \Delta_x b = 0 \Rightarrow S(O, (\pi(x), b)) = b. \end{aligned}$$

The public-data-only contract is

$$\exists S, \quad \forall O, \quad \mathsf{CVP}_{\mathrm{corr}}(O) \Rightarrow \big(\mathsf{H}_O(S) \wedge \mathsf{F}_O(S)\big).$$

Proposition 3.4 (Typed HCP ambiguity and CVP oracle insufficiency)**.** *Let $x, y \in A$ be representatives in a typed HCP semantic realization. Suppose that M_x and M_y are homotopy equivalent and that some $b \in \mathbb{R}^n$ satisfies*

$$\Delta_x b = 0, \qquad \Delta_y b \neq 0.$$

If at least one correct explicit-lattice CVP oracle exists, then

$$\neg\exists S, \quad \forall O, \quad \mathsf{CVP}_{\mathrm{corr}}(O) \Rightarrow (\mathsf{H}_O(S) \wedge \mathsf{F}_O(S)).$$

Hence a correct oracle for explicit known-lattice CVP does not by itself determine the operative Hodge–Laplacian kernel.

Proof. By Proposition 2.13, $\pi(x) = \pi(y)$. Suppose $\exists O, \mathsf{CVP}_{\mathrm{corr}}(O)$, and choose such an O. Suppose, for contradiction, that S satisfies the displayed public-data-only contract. Then

$$\mathsf{H}_O(S) \wedge \mathsf{F}_O(S).$$

Since $\Delta_x b = 0$, exactness at x gives

$$S(O, (\pi(x), b)) = b.$$

Soundness at y gives

$$\Delta_y S(O, (\pi(y), b)) = 0.$$

Replacing $\pi(y)$ by $\pi(x)$ and using the exactness equality yields $\Delta_y b = 0$, contradicting the separation hypothesis. □

For every dimension at least two, consider the elementary matrix witness $L = 0$, $L' = \mathrm{diag}(1, 0, \ldots, 0)$, and $b = e_1$ under the constant projection to a one-point public type. This is a sanity example showing that the abstract ambiguity predicates are inhabited. It is not used as evidence that an arbitrary HCP realization admits the required semantic and Hodge extension witness. That witness remains an explicit hypothesis of Proposition 3.4; the following family constructs one for its stated semantic fragment.

3.5 A Constructed Integral Cellular Family

The preceding implication has a concrete instantiation in a stated homological fragment of the typed Helfer interface. Take one vertex, one edge and one face with incidence numbers $\partial_1 = 0$, $\partial_2 = 1$, and stalks $0, \mathbb{R}^2, \mathbb{R}$, respectively. This is the algebraic finite signed-incidence model of Section 2. Two cellular sheaves in that model have

$$\delta_{0,0} = \delta_{0,1} = 0, \qquad \delta_{1,0}(x_0, x_1) = x_0, \qquad \delta_{1,1}(x_0, x_1) = x_1.$$

Their true degree-one Hodge operators are

$$\Delta_0 = \mathrm{diag}(1, 0), \qquad \Delta_1 = \mathrm{diag}(0, 1).$$

Coordinate exchange is a degreewise chain isometry. Both operators have positive spectral gap 1. Standard integral stalk lattices give harmonic lattices $\mathbb{Z}e_1$ and $\mathbb{Z}e_0$, each with an explicit full-span presentation.

Interpret these degree-zero-through-two complexes in the bounded-below real cochain Quillen model category, with the incidence-defined coboundaries displayed above. The closed tests are indexed by integers and ask whether cohomology in that degree is nonzero. The required cofibrancy and fibrancy hold for these real complexes. The model's weak equivalences are quasi-isomorphisms, which preserve the tests and give the required closed-test invariance under the declared typed Helfer equivalences.

The coordinate-exchange chain isometry gives the same homological theory to both presentations. On this admitted family, equality of the public theories is equivalent to typed homotopy equivalence.

Proposition 3.5 (One observation, two nonzero integral harmonic targets)**.** *Let p be the common closed theory of these two presentations in the stated fragment, and let $b = e_0 + e_1$. There are structured Hodge samples with that same observation (p, b), harmonic messages e_1 and e_0, and zero exact components. Their operative targets are distinct and nonzero. For each presentation's explicit harmonic lattice, its message is also the unique closest lattice vector to b.*

Proof. In the first presentation use harmonic message e_1, exact opening zero and coexact opening 1, producing noise e_0. In the second use message e_0, exact opening zero and coexact opening 1, producing noise e_1. Both sums equal b, and both messages belong to their standard integral harmonic lattices. The actual Hodge projections recover those messages. Orthogonality gives the Pythagorean identity from Proposition 3.2; any other lattice point has strictly positive squared distance from the recovered message. Thus each message is the unique CVP answer in its presented lattice. □

For the same realization, the fixed coordinate e_1 is harmonic for Δ_0 and not for Δ_1. This constructs the separation witness used in Proposition 3.4. Equality of the public theory is derived from the actual semantic equivalence, rather than imposed by a one-point classifier. The matrix-level public projection is extended from the two admitted operators; no semantic realization is asserted for other matrices.

The displayed pair uses coefficient 1 in each presentation's own harmonic-lattice basis. Its distinct ambient targets therefore do not alone assert distinct encoded integer plaintexts. A second constructed pair makes the encoding explicit: take $b = e_0 + 2e_1$, harmonic messages $2e_1$ and e_0, and

complementary coexact noise e_0 and $2e_1$. Their integer coefficients in the same declared lattice presentations are 2 and 1. Both messages are nonzero, both exact components remain zero, and the public observation is again shared. No function of that observation can return the correct coefficient for both presentations. Each message remains the unique CVP answer in its actual lattice.

Corollary 3.6 (Strict separation for hidden lattice recovery)**.** *Fix a correct CVP oracle for the presented rank-one lattices of the constructed family. When the operative lattice is supplied, the oracle recovers its unique closest vector, equal to the operative harmonic target. From the shared public class and cochain alone, no solver with that same fixed oracle can recover the operative target for both hidden members. Every larger compatible family retaining those two members inherits this exact-recovery obstruction.*

Proof. For each supplied lattice, oracle correctness and Proposition 3.5 identify its answer with the harmonic message. The two hidden members have the same public observation and distinct operative targets. A solver receives the same observation and the same oracle in either case, so its single output cannot equal both targets. The argument allows arbitrary fixed auxiliary information, including all candidate CVP answers, and imposes no running-time bound. Any compatible family containing the pair retains this contradiction. □

Solving the candidate CVP instances does not select the operative HCP answer.

4 Conclusion and Further Work

An explicitly presented harmonic lattice gives a precise discrete-CVP comparison, while the constructed common-input examples prevent uniform exact recovery from the specified public data. The exact-component obstruction shows why a partial observation need not retain the full Hodge residual.

4.1 Further Questions

The comparison suggests studying encoded HCP families with explicit distributions on hidden presentations and Hodge-structured noise. Their computational difficulty must be assessed within that specified model.

Discrete Morse reduction for cellular sheaf cohomology [CGN15] suggests a computational refinement. A chain-homotopy equivalence alone does not

identify the original harmonic representative or preserve its distance. An application to the original recovery target needs explicit retained lifts and a metric comparison.

Further work includes extending the homological model to full first-order fibrational semantics, realizing richer families topologically, and studying transported observations.

A Statement–Lean Parity Tables

Tables 2 to 4 record the correspondence for this paper's six Hodge/CVP groups and one observation-boundary group. The independent entry's configuration, `lean/Palomar/PaperOne/comparator.json`, selects 18 manuscript-facing and 20 supporting theorem declarations. Every declaration appears once below. These are coverage counts, not a count of independent research contributions. Definition rows name canonical implementation objects, not additional selected theorem targets. Canonical consequences identify derived implementation lemmas outside that configured selection. The main and supporting statements are proved under their stated premises. The final table records definitions, background, assumptions and application contracts separately.

`Palomar.PaperOne.Challenge` states the comparison objects independently using Mathlib-only imports. `Palomar.PaperOne.Solution` identifies them with the canonical development without importing Challenge. Challenge's 20 deliberate theorem placeholders are comparison targets; the proved Solution has no holes.

Write `HE` for `HomotopicalEncryption` and `E` for `HE.Palomar.PaperOne`, `CS` for `E.CellularSheaf`, `HB` for `E.HCPBridge`, and `AF` for `E.AxisFamily`. The abbreviated anchors expand to the exact configured declaration names.

Table 2: Main statement groups

Statement and scope	Formal declarations or boundary
P1-S01. Proposition 2.7; private recovery in Sections 2–3. The complex equation and actual adjoint Hodge operator give componentwise unique harmonic, exact and coexact parts. Integer coefficient recovery is classical and noncomputable in the supplied integral harmonic lattice.	`E.Hodge_Decomposition_Theorem` `E.Hodge_Decomposition_unique` `CS.HodgeEncryptedCochain.harmonicOneProjection_observed` `CS.HarmonicLatticePresentation.decodeCoefficients_encrypt`
P1-S02. Proposition 3.1. A field-linear image of the displayed generator is realized as a kernel. The consequence concerns that image, not integer-lattice membership.	`E.exists_matrix_ker_eq_range` `E.Hodge_GeneratedSubspace_Reduction_Proof`
P1-S03. Proposition 3.2. The discrete lattice is explicitly supplied and lies in the projected subspace. The Pythagorean identity preserves exactly its closest-vector answers.	`E.squaredDistance_projection_decomposition` `E.solution_iff_projectedTarget`
P1-S04. Cellular specialization of Proposition 3.2. For the actual Hodge sample and supplied harmonic message lattice, the operative message is the unique closest lattice point.	`E.concreteSheafHCP_CVP_solution_iff_message` `E.concreteSheafHCP_CVP_existsUnique`

Continued on the next page.

Table 2: Main statement groups (continued).

Statement and scope	Formal declarations or boundary

P1-S05. Propositions 3.4 and 3.5 and Corollary 3.6. Generic separation retains closed-test invariance and kernel separation; oracle corollaries also retain existence of a correct explicit-lattice CVP oracle. The constructed integral family has one public observation and distinct nonzero targets. Its supplied rank-one lattice is solvable by that same correct oracle; shared-class recovery is impossible, as is exact recovery in every compatible family retaining the separated pair.	`HB.operativeTargets_ne_of_kernelSeparation`
	`HB.cvpOracle_insufficient_of_kernelSeparation`
	`HB.cvpOracle_insufficient_for_operativeTarget`
	`AF.cellular_axis_ambiguity`
	`AF.cvp_oracle_insufficient`
	Canonical consequences:
	`HE.HCPBridge.no_solver_exactOnOperativeTarget_on_family`
	`HE.HCP.AxisFamily.cvp_known_member_hidden_member_separation`

P1-S06. Weighted example in Section 3.5. The common observation is $b = (1, 2)$, with private integer coefficients two and one. Each privately presented lattice has its own unique CVP answer.	`AF.weighted_axis_recovery_obstruction`
P1-B01. Proposition 2.8. An allowed nonzero coexact direction prevents the full Hodge residual from factoring through the exact component. The farness obstruction uses the unrestricted noise carrier and exact-component-only extraction.	`CS.HodgeEncryptedCochain.not_residual_norm_factors_exactComponent` `CS.HodgeEncryptedCochain.not_far_preserving_exactComponent`

Table 3: Supporting statements

Statement and scope	Formal declarations or boundary

Complexes and semantics. Sections 2–3. Actual coboundary composition, the constructed complex isomorphism and the corresponding three-term differential.	`CS.delta1_comp_delta0`
	`AF.cellular_complex_iso`
	`E.ThreeTermData.differential_comp`
	`AF.boundarySqZero`

Axis coordinates. Section 3.5. Coordinate vectors and the distinct complementary-axis choices in the examples.	`AF.unitCochain_apply`
	`AF.other_zero`
	`AF.other_one`
	`AF.other_ne`

Axis Hodge operators. Section 3.5. Both coboundaries, the adjoint, the degree-one Laplacian and its harmonicity criterion.	`AF.delta0_eq_zero`
	`AF.delta1_apply`
	`AF.delta1_adjoint_eq`
	`AF.degreeOneLaplacian_apply`
	`AF.harmonic_iff`

Integral lattice. Section 3.5. Restriction-compatible stalk integrality, harmonic basis vectors, encoding injectivity and membership in the chosen lattice.	`AF.standardStalkLattice_contains_iff`
	`AF.restriction_preserves`
	`AF.unitCochain_integral`
	`AF.other_unit_harmonic`
	`AF.harmonicBasis_injective`
	`AF.harmonicLatticeVector_eq`
	`AF.latticeVector_mem`

Table 4: Definitions and proof boundaries

Statement and scope	Formal declarations or boundary

HCP input and target. Section 2. The public pair contains a classifier value and observed cochain. A private representative selects the operative kernel and hence the orthogonal-projection target.	Canonical definitions:
	`HE.HCPBridge.Observation`
	`HE.HCPBridge.HiddenHodgeWitness`
	`HE.HCP.operativeHarmonicComponent`
	`HE.HCPBridge.operativeTarget`

Continued on the next page.

Table 4: Definitions and proof boundaries (continued).

Statement and scope	Formal declarations or boundary

Explicit-lattice CVP. Section 3.1. An injective real basis presents the lattice through integer coefficients. An instance adds a target; a solution is a closest discrete lattice point.	Canonical definitions:
	`CVP.ExplicitLattice`
	`CVP.Instance`
	`CVP.Solution`

Semantic realization. Section 2. The public classifier factors through the specified closed theory. The operative Hodge operator is separate presentation-level data; equal classifier values need not identify it. Completeness adds reflection of the declared homotopy equivalence; with invariance, public fibers are exactly its classes. The constructed family proves this condition.	Canonical definitions and classifier theorems:
	`HE.Axioms.PublicClassifier`
	`HE.Helfer.ClosedTheory`
	`HE.HCPBridge.Realization`
	`HE.HCPBridge.ReflectsHomotopyEquivalence`
	`HE.HCPBridge.samePublicClass_iff_homotopyEquivalent`
	`HE.HCPBridge.publicClassOf_eq_setOf_homotopyEquivalent`
	`HE.HCP.AxisFamily.theory_eq_iff_homotopyEquivalent`

Hodge sample and lattice. Sections 2–3. The sample combines an integral harmonic message with exact/coexact noise. A supplied harmonic-lattice presentation has full real span; it need not contain every intrinsic integral harmonic cochain.	Canonical definitions:
	`HE.Sheaf.CellularSheaf.HodgeEncryptedCochain`
	`HE.Sheaf.CellularSheaf.HarmonicLatticePresentation`

External theorem and semantic premise. Sections 2.2 and 3.5; distinct from the proved homological fragment.	Helfer's general fibrational/path-invariance theorem is cited background. The generic bridge uses the declared closed-test invariance; the concrete family proves it for its homological tests. Full first-order invariance is not a theorem of this entry.

Continued on the next page.

Table 4: Definitions and proof boundaries (continued).

Statement and scope	**Formal declarations or boundary**
Assumption: computational hardness. Section 3.1; distinct from the proved exact comparisons.	Known-lattice CVP hardness is an external computational interface. The selected algebraic comparisons and non-identifiability examples do not establish an asymptotic encoded-instance hardness reduction or a sampled encryption-security theorem.
Contract: protocol application. Section 2.2; distinct from the observation theorems.	A larger released view requires its own equality or factorization proof. Artifact binding, encoding, visibility and source-to-code metric comparisons are additional application conditions. No general full-HCP distance comparison follows from an exact-component extraction.

The checked declarations use only `propext`, `Quot.sound` and `Classical.choice`, or subsets. Verification applies to the exact source and configuration checked.

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